

Figure 2

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Contents

1	Packages, data and functions	1
1.1	Packages	1
1.2	Data	1
1.3	Functions	2
2	Figure 2	2
2.1	Models	2
2.2	Figure	4
3	Versions of R and of the R-packages used	11

1 Packages, data and functions

1.1 Packages

```
library(tidyverse)
library(patchwork)
library(here)
library(lme4)
library(lmerTest)
```

1.2 Data

```
load(here("Data/3_ForecastsResults/3_MergedResults/complete.forecasts.RData"))

complete.forecasts.taxa <- complete.forecasts %>%
  dplyr::filter(
```

```

forecast.method == "Simplex EDM",
Target != "Total_biomass",
Target != "Mean_oxygen"
)

```

1.3 Functions

```

conf_interval <- function(model, nd, response, renameResponse, cmult=1.96){
  nd[,response] <- predict(model, nd, re.form=NA)
  mm <- model.matrix(terms(model),nd)
  pvar1 <- diag(mm %*% tcrossprod(vcov(model),mm))
  nd <- nd %>%
    dplyr::mutate(lower=!!as.name(response)-cmult*sqrt(pvar1),
                  upper=!!as.name(response)+cmult*sqrt(pvar1),
                  Response=renameResponse) %>%
    dplyr::rename(fit=!!as.name(response))
  return(nd)
}

```

2 Figure 2

2.1 Models

```

## acf vs exp
m.acf.exp <-
  lmer(
    acf_lag5 ~ median_richness_center * light_treatment +
      (1 |Target) + (1 | bottle) + (1 | incubator),
    complete.forecasts.taxa
  )

## cv vs exp
m.cv.exp <-
  lmer(

```

```

    cv.res.log10 ~ median_richness_center * light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    complete.forecasts.taxa
  )

### pe vs rmse
m.pe.exp <-
  lmer(
    pe5 ~ median_richness_center * light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    complete.forecasts.taxa
  )

## acf vs RMSE
m.acf.rmse <-
  lmer(
    RMSE ~ acf_lag5 * light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    complete.forecasts.taxa
  )

## CV vs RMSE
m.cv.rmse <-
  lmer(
    RMSE ~ cv.res.log10 * light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    complete.forecasts.taxa
  )

### pe vs rmse
m.pe.rmse <-
  lmer(
    RMSE ~ pe5 * light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    complete.forecasts.taxa
  )

```

2.2 Figure

```
newdat.m.exp <-  
  expand.grid(  
    median_richness_center = seq(  
      min(complete.forecasts.taxa$median_richness_center),  
      max(complete.forecasts.taxa$median_richness_center),  
      length.out = 100  
    ),  
    light_treatment = unique(complete.forecasts.taxa$light_treatment)  
  )  
  
newdat.m.acf <-  
  expand.grid(  
    acf_lag5 = seq(  
      min(complete.forecasts.taxa$acf_lag5),  
      max(complete.forecasts.taxa$acf_lag5),  
      length.out = 100  
    ),  
    light_treatment = unique(complete.forecasts.taxa$light_treatment)  
  )  
  
newdat.m.cv <-  
  expand.grid(  
    cv.res.log10 = seq(min(log10(  
      complete.forecasts.taxa$cv.res  
    )),  
    max(log10(  
      complete.forecasts.taxa$cv.res  
    )),  
    length.out = 100),  
    light_treatment = unique(complete.forecasts.taxa$light_treatment)  
  )  
  
newdat.m.pe.rmse <-  
  expand.grid(  
    
```

```

pe5 = seq(
  min(complete.forecasts.taxa$pe5),
  max(complete.forecasts.taxa$pe5),
  length.out = 100
),
light_treatment = unique(complete.forecasts.taxa$light_treatment)
)

## acf vs exp
newdat.acf.exp <-
  conf_interval(m.acf.exp, newdat.m.exp, "acf_lag5", "Autocorrelation at lag 5")

## cv vs exp
newdat.cv.exp <-
  conf_interval(m.cv.exp, newdat.m.exp, "cv.res.log10", "cv.res.log10")

## acf vs RMSE
newdat.m.acf.rmse <-
  conf_interval(m.acf.rmse, newdat.m.acf, "RMSE", "Forecast error (RMSE)")

## CV vs RMSE
newdat.m.cv.rmse <-
  conf_interval(m.cv.rmse, newdat.m.cv, "RMSE", "Forecast error (RMSE)")

### pe vs exp design
newdat.m.exp <-
  conf_interval(m.pe.exp, newdat.m.exp, "pe5", "Permutation entropy")

### pe vs rmse
newdat.m.pe.rmse <-
  conf_interval(m.pe.rmse, newdat.m.pe.rmse, "RMSE", "Forecast error (RMSE)")

fig2exp.acf <- complete.forecasts.taxa %>%
  ggplot(aes(median_richness_center, acf_lag5, col = light_treatment)) +

```

```

geom_ribbon(
  data = newdat.acf.exp,
  aes(
    ymin = lower,
    ymax = upper,
    y = fit,
    group = light_treatment
  ),
  col = NA,
  fill = "grey",
  alpha = 0.3
) +
geom_line(data = newdat.acf.exp,
  size = 1.3,
  aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
labs(x = "Centered median realized species richness",
  y = "Autocorrelation at lag 5",
  col = "Light conditions") +
theme_bw() +
theme(legend.position = "none")

fig2exp.cv <- complete.forecasts.taxa %>%
  ggplot(aes(median_richness_center, cv.res.log10, col = light_treatment)) +
  geom_ribbon(
    data = newdat.cv.exp,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,
    fill = "grey",
    alpha = 0.3
  ) +

```

```

geom_line(data = newdat.cv.exp,
          size = 1.3,
          aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
labs(x = "Centered median realized species richness",
     y = "log10(Coefficient of variation CV)",
     col = "Light conditions") +
theme_bw() +
theme(legend.position = "bottom")

fig2acf.rmse <- complete.forecasts.taxa %>%
  ggplot(aes(acf_lag5, RMSE, col = light_treatment)) +
  geom_ribbon(
    data = newdat.m.acf.rmse,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,
    fill = "grey",
    alpha = 0.3
  ) +
  geom_line(data = newdat.m.acf.rmse,
            size = 1.3,
            aes(y = fit, group = light_treatment)) +
  geom_point(size = 2.5, alpha = 0.6) +
  labs(x = "Autocorrelation at lag 5", y = "Forecast error (RMSE)",
       col = "Light conditions") +
  theme_bw() +
  theme(legend.position = "none")

fig2cv.rmse <- complete.forecasts.taxa %>%
  ggplot(aes(cv.res.log10, RMSE, col = light_treatment)) +
  geom_ribbon(

```

```

data = newdat.m.cv.rmse,
aes(
  ymin = lower,
  ymax = upper,
  y = fit,
  group = light_treatment
),
col = NA,
fill = "grey",
alpha = 0.3
) +
geom_line(data = newdat.m.cv.rmse,
  size = 1.3,
  aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
labs(x = "log10(Coefficient of variation CV)",
  y = "Forecast error (RMSE)",
  col = "Light conditions",
  tag = "e") +
theme_bw() +
theme(legend.position = "none")

fig2exp.pe <- complete.forecasts.taxa %>%
  dplyr::filter(Target != "Total_biomass",
    Target != "Mean_oxygen",
    forecast.method == "Simplex EDM") %>%
  ggplot(aes(median_richness_center, pe5, col = light_treatment)) +
  geom_ribbon(
    data = newdat.m.exp,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,

```

```

    fill = "grey",
    alpha = 0.3
) +
geom_line(data = newdat.m.exp,
          size = 1.3,
          aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
labs(x = "Centered median realized species richness",
     y = "Permutation entropy",
     col = "Light conditions") +
theme_bw() +
theme(legend.position = "none")

fig2pe.rmse <- complete.forecasts.taxa %>%
  dplyr::filter(Target != "Total_biomass",
                Target != "Mean_oxygen",
                forecast.method == "Simplex EDM") %>%
ggplot(aes(pe5, RMSE, col = light_treatment)) +
geom_ribbon(
  data = newdat.m.pe.rmse,
  aes(
    ymin = lower,
    ymax = upper,
    y = fit,
    group = light_treatment
  ),
  col = NA,
  fill = "grey",
  alpha = 0.3
) +
geom_line(data = newdat.m.pe.rmse,
          size = 1.3,
          aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
labs(y = "Forecast error (RMSE)", x = "Permutation entropy",
     col = "Light conditions") +

```

```
theme_bw() +
theme(legend.position = "none")
```

```
fig2acf.rmse + fig2cv.rmse + fig2pe.rmse +
fig2exp.acf + fig2exp.cv + fig2exp.pe +
plot_layout(ncol = 3) +
plot_annotation(tag_levels = "a") &
scale_color_manual(values = c("#C4961A", "#293352"),
aesthetics = c("color", "fill"))
```

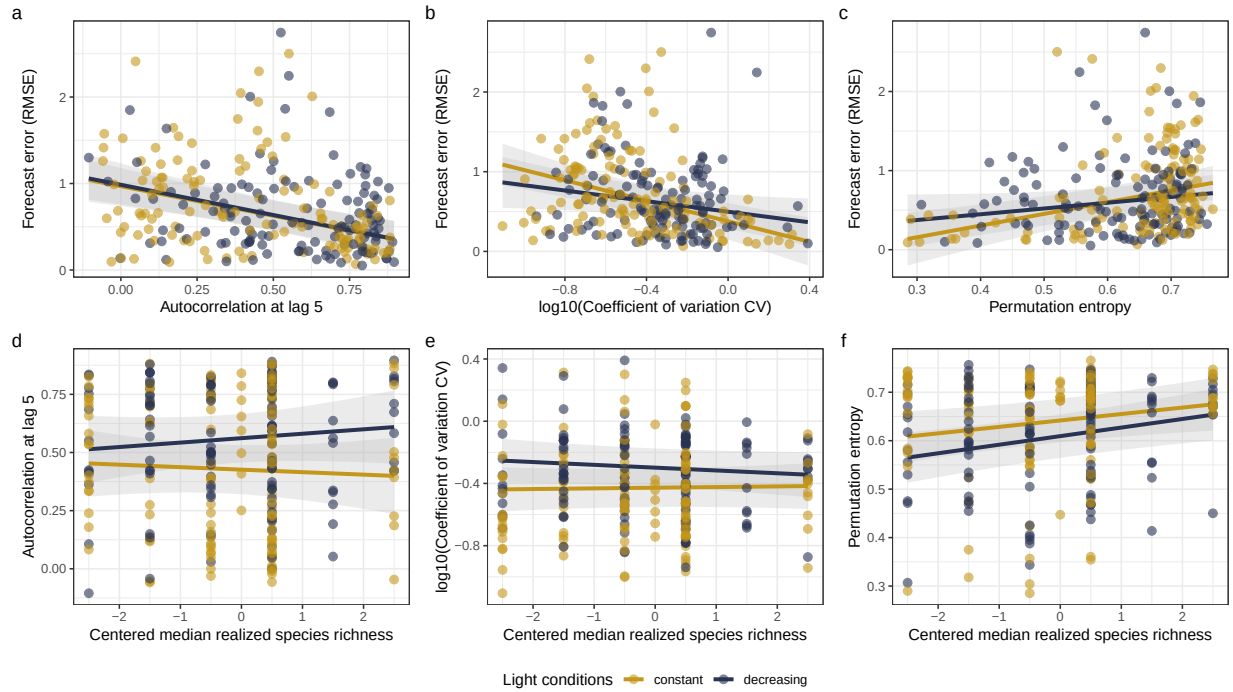


Figure 2: Relationship of time series metrics with forecast error (a-c), and with experimental treatments (d-f). Time series metrics are (a, d) autocorrelation at lag 5, (b, e) coefficient of variation as a measure of temporal variability, and (c, f) permutation entropy as a measure of time series complexity. Lines display the fit of the corresponding linear mixed model, shaded areas denote the 95% confidence intervals. Taxon, bottle, and incubator were included as random effects in the model. Forecast error is based on forecasts computed with Simplex EDM.

3 Versions of R and of the R-packages used

R version: R version 4.3.2 (2023-10-31 ucrt).

Base packages: stats, graphics, grDevices, utils, datasets, methods, base.

Other packages: lmerTest 3.1-3, lme4 1.1-35.1, Matrix 1.6-5, here 1.0.1, patchwork 1.2.0, lubridate 1.9.3, forcats 1.0.0, stringr 1.5.1, dplyr 1.1.4, purrr 1.0.2, readr 2.1.5, tidyr 1.3.1, tibble 3.2.1, ggplot2 3.4.4, tidyverse 2.0.0, knitr 1.45.

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